PanNuke: final experiment report

Three questions: block fidelity, cost scaling, sibling structure

Method	Bio mIoU	Delta (pp)	W/L/T
Raw TP	0.472126	+0.0000	-
V1: no block fidelity	0.472389	+0.0262	58/34/8
V2: block fidelity	0.472389	+0.0263	58/34/8
BCSS scaling: matched	0.472407	+0.0280	56/36/8
BCSS scaling: dev-selected	0.472228	+0.0101	55/36/9
Parent-child only	0.472262	+0.0136	57/35/8
Generic Potts siblings	0.472480	+0.0354	58/34/8

• V1 and the biological-sibling reference are exactly the same prediction. All rows use the same 100 full-patch comparison inputs.

• Main result: generic Potts is best numerically. Biological siblings do not establish added value over it. Some patches worsen under every tested WCSP.

Method and experimental controls

• HoVer-Net raw six-channel TP softmax provides pixel evidence. Labels: Background, Neoplastic, Inflammatory, Connective, Dead, Non-Neoplastic Epithelial.

• Six nested levels cover 1, 2, 4, 8, 16 and 32 pixel blocks. Blocks have soft dominant-class states. Root is a container, not a variable.

• Pixel unary = -log probability, shifted to minimum zero. Parent-child mismatch and same-level adjacency add weighted costs.

Group	Controlled change
Block fidelity	V1: no block unary. V2: mean-probability block unary, gamma .3.
Scaling	Block unary x area; sibling x side; mismatch x child area. Matched parent .1; dev-selected parent .01.
Sibling ablation	Original unscaled V1, gamma 0, parent .1, sibling .3. Only sibling tables differ: zero / Potts / hybrid.

• Tables preserve the previously specified revised biological prior and scale-specific learned statistics, blended 20% prior / 80% learned. Potts and hybrid off-diagonal means both equal one.

• Bidirectional strict-decrease ICM, at most 100 rounds; independent energy and local-move audits. No global optimum claimed.

Data and statistical interpretation

• Reuse 200 fitting patches, 12 separate development patches and the original 100 comparison patches. No new inference or optimization was run during final reporting.

• Full 256x256 predictions use four overlapping 256-input / 164-output tiles with 46-pixel reflect padding and averaged overlap probabilities. No evaluation crop or resizing.

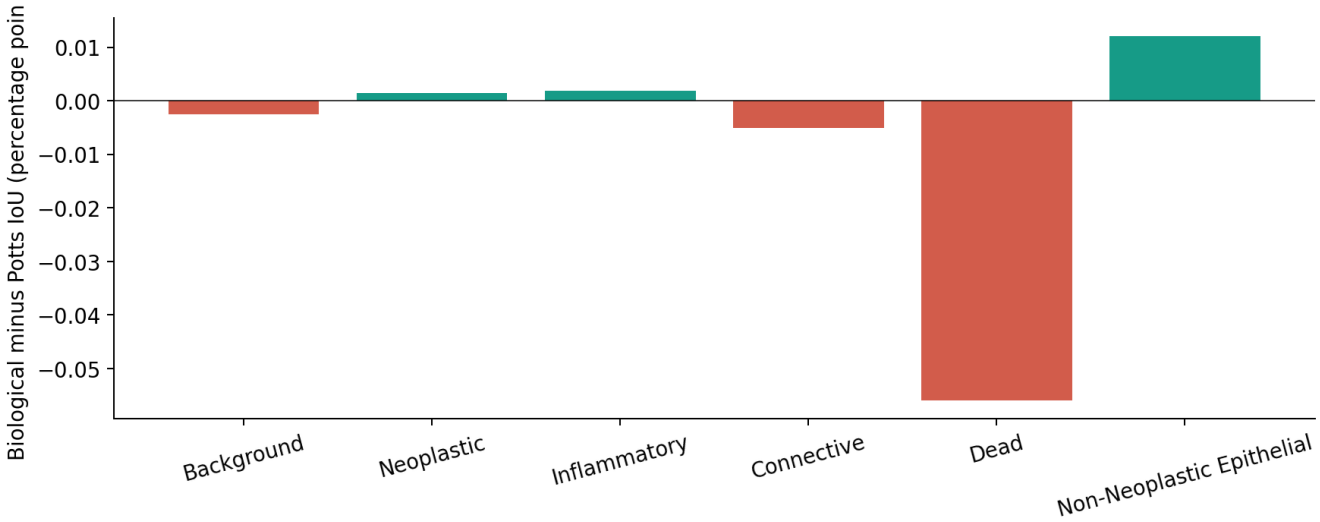
• Comparison is exploratory: patches were already inspected, and four IDs overlap a later-fetched upstream beta-development sweep (908, 2216, 2330, 2354). Patient/slide separation and checkpoint exposure are unverified.

• Biological mIoU pools TP/FP/FN across patches, then averages the five class IoUs. Six-class mIoU includes Background. Zero-union IoU is zero. Paired bootstrap resamples whole patches 5,000 times.

Difference	Delta (pp)	95% paired CI (pp)
Scaling matched - V2	+0.0017	[-0.0054, +0.0087]
Scaling selected - V2	-0.0162	[-0.0315, -0.0017]
Parent-child - raw	+0.0136	[-0.0042, +0.0299]
Potts - parent-child	+0.0218	[+0.0020, +0.0367]
Biological - Potts	-0.0091	[-0.0164, +0.0022]

• An interval containing zero does not establish a positive effect. Class-level point estimates alone do not demonstrate biological validity.

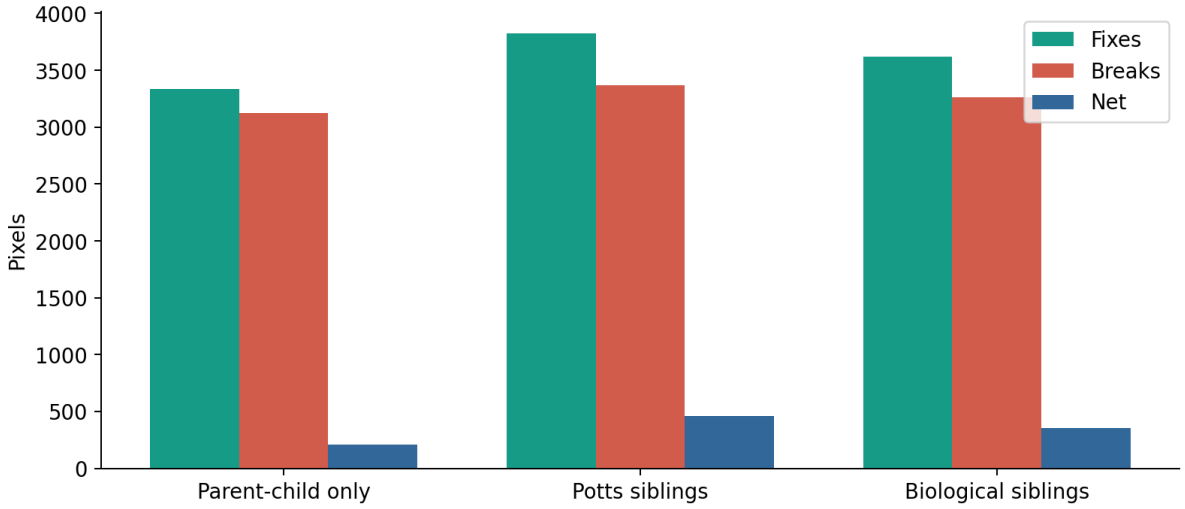
Sibling constraints: class effects



Class	Raw	Parent-child	Potts	Biological
Background	0.94666	0.94667	0.94669	0.94667
Neoplastic	0.63326	0.63347	0.63362	0.63364
Inflammatory	0.44336	0.44340	0.44331	0.44333
Connective	0.47591	0.47592	0.47602	0.47597
Dead	0.33322	0.33350	0.33433	0.33377
Non-Neoplastic	0.47487	0.47502	0.47512	0.47524

• Biological minus Potts: Dead loses most; Non-Neoplastic gains slightly. These are descriptive effects without class-specific uncertainty tests.

Where corrections occur

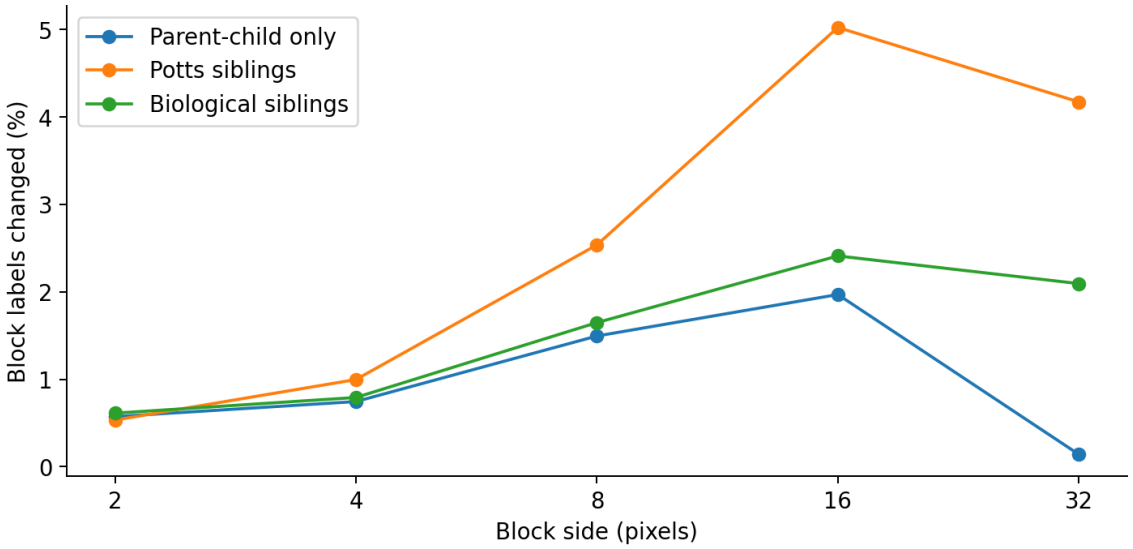


Condition	Boundary fixes / breaks	Interior fixes / breaks
parent only	1850/1855	1484/1270
potts	2015/2055	1812/1314
biological	1939/1966	1678/1297

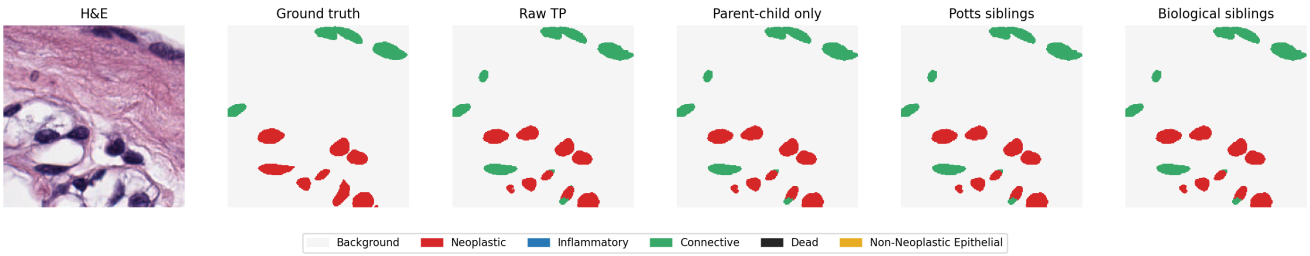
• Boundary means within two pixels of a ground-truth label transition. Most changed predictions are uncertain: fixed/broken top-two margins are around .02-.03, versus .978 for unchanged correct pixels.

• All five new conditions reached audited local minima. Solver-plus-audit times are about 6.6-7.1 seconds per condition across 100 patches; inference was reused.

Hierarchy behavior and examples



Fold 2 / 2015 - Kidney; biological minus Potts patch mIoU +0.000000



• Block-label changes are greatest near 16x16 in these runs. That does not prove a level is necessary or unnecessary; a level-removal experiment would be needed.

• The representative biological-versus-Potts example is selected near the median effect. Favorable and harmful extremes are also saved in figures/ and the slide deck.

Conclusions and next questions

• Block fidelity adds essentially no reliable advantage: only 83 labels differ between V1 and V2 across 6,553,600 pixels.

• BCSS-style scaling does not establish a benefit on PanNuke. Matched-setting change is inconclusive; the development-selected variant loses to unscaled V2.

• Potts contributes beyond parent-child consistency in the reported sample. The biological table does not establish an advantage over Potts.

• The no-degradation target remains unmet: parent-child worsens 35 patches; Potts and biological each worsen 34. Never select per-patch fallbacks using evaluation ground truth.

• PanNuke annotates nuclei; Background dominates many mixed blocks. Dead appears in only two fitting patches and no 32x32 fitting block has dominant Dead.

• Possible next designs: retain class proportions, use class-sensitive parent-child penalties, and validate rules at their intended scales. Any chosen revision needs frozen development choices and new comparison patches.

• BCSS includes convergence-time controls, oracle/error-location ceilings and class-presence analyses not replicated here. They are research options, not results to claim.

Reproducibility

• REPORT.md contains detailed tables, confidence and boundary summaries, examples, commands and limitations. summary.json, comparisons.json, manifests, frozen tables/configs, compact confusion arrays and bootstrap draws retain the evidence.

• Original V1/V2 and subsequent outputs remain unchanged locally. Publication excludes raw images, checkpoint, probabilities, environments, internal guides, build files, logs and bulky per-patch arrays.